

Chapter 54 — Management of Paediatric Kidney Stone Disease — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Ultrasound (US) first**; low-dose non-contrast CT (NCCT) only if US non-diagnostic + high suspicion.
- **Every stone-forming child → full metabolic evaluation**: stone analysis + 24-h urine.
- **Commonest abnormality: hypercalciuria (30–50%)**, mostly idiopathic; >4 mg/kg/day.
- **Uncomplicated ureteric stone ≤ 10 mm → observation $\pm$ α -blocker** (off-label, ≤ 6 weeks).
- **Renal ≤ 20 mm → SWL or URS first-line** (shockwave lithotripsy, ureteroscopy).
- **Renal >20 mm → PCNL** (percutaneous nephrolithotomy), clearance about 90%.
- **No routine pre-stenting** before URS.
- **Low-dose CT before PCNL** — strong recommendation, grade C evidence.
- **Do not restrict calcium; restrict sodium**; fluid >2 – 2.5 L/day (adolescents).
- **PH1 (primary hyperoxaluria type 1) → pyridoxine trial in all; lumasiran FDA-approved.**

EPIDEMIOLOGY

- **Recurrence** about 50% within 5 years (3–5 years in children).
- **US prevalence** 5% → 11%; youth incidence +5–10%/year.
- **Females 15–19 y**: 75 → 120/100,000; 52% > males.
- **Sex ratio**: adult M : F 3 : 1 vs **early-onset female-predominant**.
- **Early-onset** = onset <24 years.
- **Commonest stone**: calcium oxalate (CaOx) about 80%, as in adults; children more CaP, less uric acid (alkaline urine).
- **Randall's plaque** = papillary **hydroxyapatite**; platform for idiopathic CaOx.
- **Antibiotic-linked stones**: risk persists up to 5 years; greatest in children.
- **Drug-crystal stones** 1–2%.
- **Spina bifida post-augmentation**: about 1%/year, $\geq 10 \times$ baseline.
- **Fracture risk peaks** at 10–19 years.

EVALUATION

- **Gold standard**: NCCT, about 100% sensitivity/specificity.
- **US**: $>70\%$ sensitive, $>95\%$ specific; ureteric stones poorly seen.
- **US stone** = hyperechoic focus + **twinkle**; **maximise PRF** (pulse repetition frequency).
- **No shadow** for stones <4 mm; US overestimates size 1–2 mm.
- **Low-dose CT**: 40 mA for ≤ 50 kg.
- **Image Gently Alliance**: 2007.
- **Metabolic abnormality** → up to $5 \times$ recurrence.
- **Stone analysis**: infrared spectroscopy / X-ray diffraction; diagnostic — cystine, struvite, 2,8-DHA (2,8-dihydroxyadenine), xanthine.
- **24-h urine**: Ca, oxalate, urate, Na, citrate, creatinine, volume, pH, cystine.
- **Not toilet-trained** → spot solute : creatinine ratios.
- **PTH (parathyroid hormone) not routine** — only if $\uparrow$ Ca + $\downarrow$ phosphate.
- **CYP24A1**: $\uparrow$ 1,25-(OH) $_2$ D mimics vitamin D intoxication; screen 25-OH-D : 24,25-(OH) $_2$ D.
- **Dent screen**: urine β_2 -microglobulin / retinol-binding protein.
- **Crystals**: hexagonal = cystine; coffin-lid = struvite; brown spiculated = 2,8-DHA.

URINARY METABOLIC ABNORMALITIES

- **Hypercalciuria**: >4 mg/kg/day (>2 y) or >200 – 250 mg/day.
- **Spot Ca : Cr**: ≤ 0.21 school-age; <0.8 at 0–6 mo; <0.6 at 7–12 mo.
- **Idiopathic hypercalciuria**: 4% of healthy children; autosomal dominant, incomplete penetrance.

- **Hypercalciuria work-up**: serum calcium first → high / low / normal bins.
- **Oxalate**: 80–85% endogenous; normal <40 (text) / <50 (table) mg/1.73 m 2 .
- **PH markers**: PH1 glycolate · PH2 glycerate · PH3 4-hydroxy-2-oxoglutarate.
- **PH genes**: AGXT (AGT, peroxisomal) · GRHPR · HOGA1 (mitochondrial).
- **All PH markers normal** + $\uparrow$ oxalate → secondary hyperoxaluria.
- **Hypocitraturia**: <128 mg/g (M), <300 mg/g (F); table male normal >180 .
- **Cystinuria**: SLC3A1/SLC7A9; normal <60 , homozygote >400 mg/day/1.73 m 2 .
- **Uric acid**: FE (fractional excretion) 30–50% neonate → 8–12%; stones $<5\%$.
- **Hyperuricosuria**: >815 mg/day/1.73 m 2 ; spot >0.56 mg/dL GFR (>2 y).
- **Uric acid stone risk**: hyperuricosuria + low urine pH.
- **B6 deficiency mimics PH1**; cystinuria can present in infancy (staghorn).
- **Hereditary renal hypouricaemia**: SLC22A12/SLC2A9; exercise-induced renal failure.
- **Eponyms**: **Dent** — X-linked CLCN5, low-molecular-weight proteinuria. **Bartter** — metabolic alkalosis. **Lesch-Nyhan** — HPRT deficiency, X-linked. **Williams** — dysmorphism, hypercalcaemic hypercalciuria.
- **X-linked stone genes**: Dent, HPRT, PRPS → boys.

MANAGEMENT — MET AND GUIDELINES

- **MET** (medical expulsive therapy) = $\alpha 1A/\alpha 1D$ blockade; **tamsulosin**.
- **Spontaneous passage**: best <5 mm, distal.
- **Up to 60%** of children need surgery.
- **AUA/EUS 2016** (American Urological Association/Endourological Society): 8 paediatric recommendations, half expert opinion.
- **Failed observation/MET** → URS or SWL (strong).
- **Asymptomatic non-obstructing** → surveillance with US.
- **No routine open/laparoscopic/robotic surgery** unless anatomic abnormality.
- **Prophylaxis**: all URS/PCNL; SWL if high-risk; stop ≤ 24 h.
- **Residual fragment cut-point**: 4 mm on US (2 mm overestimate + 2 mm CT).
- **Radiation limits**: 50 mSv/12 months; <20 mSv/year over 5 years.

URETEROSCOPY

- **Clearance** approaching 90%; **<6 y or <20 kg** → more complications.
- **Dusting**: >50 Hz, <0.4 J → <1 mm; fragmentation → 1–4 mm.
- **Scopes**: semirigid 4.5–6 Fr; flexible 7.5–9 Fr, 270° deflection.
- **4 Fr catheter / 3.7 Fr stent** → 0.025-inch wire only.
- **8/10 Fr won't pass** → 4.6–4.8 Fr stent ≥14 days.
- **Fibres**: 150–200 µm lower pole; 365 µm elsewhere.
- **Stent on string** 5–7 days.
- **Stent if**: long procedure, many passes, trauma/oedema.
- **Stricture**: distal → reimplant; proximal → pyeloplasty; mid → excision + anastomosis.

SHOCKWAVE LITHOTRIPSY

- **First paediatric SWL**: 1986.
- **Favourable**: <10 mm, <1000 HU (Hounsfield units), non-lower-pole.
- **Unfavourable**: cystine, brushite; infundibulopelvic angle >45°.
- **Settings**: <20 kV; ≤2500–3000 shocks (<2000 very young); 80 > 120/min.
- **Ungated shocks** safe in children.
- **Absolute CI** (contraindications): bleeding diathesis, untreated UTI (urinary tract infection), pregnancy.
- **SWL >20 mm** → stent or nephrostomy (AUA); authors: **no SWL >15 mm**.
- **URS = SWL** for clearance; SWL fewer anaesthetics and ED visits.
- **Steinstrasse** — ureteric column of post-SWL fragments.

PERCUTANEOUS NEPHROLITHOTOMY

- **First-line >20 mm**; clearance about 90%.
- **Child 24–30 Fr ≈ adult 72 Fr**; mini <20–24 Fr (1998); micro <10 Fr.
- **Culture** 2–3 weeks pre-op; bladder urine ≠ pelvic/stone culture.
- **Sheath** 18 Fr (24 Fr staghorn); nephrostomy 7 Fr infant → 16 Fr adolescent.
- **Lower-pole posterior calyx**: one above most inferior on AP (>90%).
- **Upper pole**: below 11th rib, medial calyx.
- **Complications**: intraoperative 10%, postoperative 20%; ≥Clavien III ≤16%; transfusion <10%.
- **Operate only** where paediatric interventional radiology can embolise.
- **Tubeless PCNL** not established in children.
- **Complication drivers**: operative time, sheath size, mid-calyceal puncture, partial staghorn.

SECONDARY PREVENTION

- **Fluid**: ≥ maintenance (child); >2–2.5 L (adolescent); 1.5–2 L/m² cystinuria/PH.
- **Sodium**: <2–3 mEq/kg/day (child); <2.4 g/day (adolescent).
- **Protein**: 100% of daily allowance; avoid excess animal protein.
- **Hydrochlorothiazide** 1–2 mg/kg/day; add potassium citrate.
- **Potassium citrate** 2–4 mEq/kg/day.
- **Allopurinol** 4–10 mg/kg/day (max 300 mg).
- **Pyridoxine** 2–5 → 8–10 mg/kg/day; 10–30% of PH1 respond.
- **pH targets**: uric acid >6.5; cystine >7; CaP risk >6.5.
- **FDA-approved (children)**: thiols + lumasiran only.
- **Commonest thiol**: tiopronin.
- **Sodium citrate** less ideal — sodium → calciuria.
- **Grapefruit** ↑ risk; orange juice, lemonade ↓ risk.
- **Stop vitamin C** in hyperoxaluric CaOx formers.
- **ILLUMINATE-A** (lumasiran): oxalate ↓ 65%; 84% ≤1.5 × ULN.

- **PUSH trial**: incentives + coaching for hydration.
- **Elastosis perforans serpiginosa** — D-penicillamine.

CLASSIC TRAPS

- **Trap**: childhood stones mostly genetic — **Actually**: mostly idiopathic CaOx.
- **Trap**: no shadow = no stone — **Actually**: stones <4 mm rarely shadow; use twinkle.
- **Trap**: check PTH routinely — **Actually**: only if ↑ Ca + ↓ phosphate.
- **Trap**: low-calcium diet — **Actually**: raises urinary oxalate; don't restrict.
- **Trap**: "strong" = strong evidence — **Actually**: CT before PCNL is grade C.
- **Trap**: pre-stent routinely — **Actually**: not recommended; more ED visits.
- **Trap**: balloon-dilate the ureter — **Actually**: passive stent dilatation only.
- **Trap**: allopurinol first for uric acid stones — **Actually**: fluid + alkali.
- **Trap**: genotype predicts pyridoxine response — **Actually**: trial in all PH1.
- **Trap**: more alkali always better — **Actually**: pH >6.5 → CaP stones.